

04_telemetry_gaps

Propietario	© Carlos Roberto
Etiquetas	

Before querying 15M rows, this section validates the telemetry dataset. Total records, date range, distinct vehicles transmitting, and active vehicles in the master table. Establishes the working baseline before any gap analysis is performed.

```
-- =====
-- IMPORTANT: Keep this block at the top of every execution session.
-- If running sections individually, declare these variables first or queries will fail.
-- =====

USE TRANSTRACK;
GO

DECLARE @window_start DATETIME2 = DATEADD(DAY, -90, GETDATE());
DECLARE @window_end DATETIME2 = GETDATE();
DECLARE @signal_interval_min INT = 5
```

4.0 TELEMETRIA_GPS size and date range

```
-- -----
-- SECTION 0: Table size and date range sanity check
-- Always check before running 50M row queries
```

```

--
--

SELECT 'SECTION 0 - TELEMETRIA_GPS size and date range' AS diagnostic_section;

```

```

SELECT
    COUNT(*) AS total_gps_records,
    MIN(fecha_hora) AS earliest_record,
    MAX(fecha_hora) AS latest_record,
    COUNT(DISTINCT vehiculo_id) AS distinct_vehicles_in_telemetry,
    (SELECT COUNT(*) FROM flota.VEHICULO WHERE estado = 'ACTIVO') AS active_vehicles_in_master,
    -- Records in our working window
    SUM(CASE WHEN fecha_hora >= @window_start THEN 1 ELSE 0 END) AS records_in_90d_window
FROM flota.TELEMETRIA_GPS;

```

diagnostic_section						
1	SECTION 0 - TELEMETRIA_GPS size and date range					
	total_gps_records	earliest_record	latest_record	distinct_vehicles_in_telemetry	active_vehicles_in_master	records_in_90d_window
1	15000000	2021-01-01 00:00:00.0000000	2025-12-30 23:55:00.0000000	800	714	0

Query executed successfully. c13\lab_db (16.0 RTM) | carlos_roberto (63) | TRANSTRACK | 00:00:01 | Row: 1, Col: 1 | 2 rows

4.1 Ghost vehicles: in telemetry, not in VEHICULO master

```

--
--
-- SECTION 1: Ghost vehicles – transmitting in GPS but not in

```

```

flota.VEHICULO
-- TELEMETRIA_GPS.vehiculo_id has no FK – any integer can app
ear here
-----

SELECT 'SECTION 1 - Ghost vehicles: in telemetry, not in VEHI
CULO master' AS diagnostic_section;

SELECT
    t.vehiculo_id                AS ghost_vehicle_id,
    COUNT(*)                     AS gps_record_count,
    MIN(t.fecha_hora)            AS first_signal,
    MAX(t.fecha_hora)            AS last_signal,
    AVG(t.velocidad_kmh)         AS avg_speed_kmh,
    MAX(t.velocidad_kmh)         AS max_speed_kmh,
    -- Is this vehicle moving? Could be sensor error or unaut
horized device
    CASE
        WHEN AVG(t.velocidad_kmh) > 60 THEN 'ACTIVELY_MOVING_
NO_RECORD'
        WHEN AVG(t.velocidad_kmh) > 0  THEN 'OCCASIONAL_MOVEM
ENT_NO_RECORD'
        ELSE 'STATIONARY_UNKNOWN_DEVICE'
    END                          AS risk_classification
FROM flota.TELEMETRIA_GPS t
LEFT JOIN flota.VEHICULO v ON t.vehiculo_id = v.vehiculo_id
WHERE t.fecha_hora >= @window_start
    AND v.vehiculo_id IS NULL      -- Not in master table
GROUP BY t.vehiculo_id
ORDER BY gps_record_count DESC;

```

Section 4.1 and 4.2 returned no results - telemetry vehicle IDs were generated within the same range as `flota.VEHICULO`, preventing ghost vehicle detection in this

environment. In production, this query would identify devices transmitting with unregistered IDs.

4.2 GPS records with no assignment overlap

```
-- -----  
--  
-- SECTION 2: GPS records with no assignment overlap  
-- For each GPS point, try to find a matching ASIGNACION_VEHICULO window.  
-- If none found, the vehicle was moving without a formal mission.  
-- NOTE: ASIGNACION.pedido_id is nullable – assignment may exist without an order.  
-- -----  
--  
  
SELECT 'SECTION 2 - GPS signals with no matching assignment  
(sample 90 days)' AS diagnostic_section;  
  
-- Aggregate by vehicle to avoid row-by-row output of 50M records  
SELECT  
    t.vehiculo_id,  
    v.placa,  
    v.estado AS vehicle_status,  
    COUNT(*) AS total_gps_points_in_window,  
    -- Points with no overlapping assignment  
    SUM(CASE WHEN av.asignacion_id IS NULL THEN 1 ELSE 0 END)  
AS unassigned_gps_points,  
    CAST(  
        100.0 * SUM(CASE WHEN av.asignacion_id IS NULL THEN 1  
ELSE 0 END)  
/ NULLIF(COUNT(*), 0)
```

```

        AS DECIMAL(5,2))                AS pct_unassigned,
        SUM(CASE WHEN av.asignacion_id IS NULL AND t.velocidad_kmh
h > 60 THEN 1 ELSE 0 END) AS high_speed_unassigned,
        MAX(CASE WHEN av.asignacion_id IS NULL THEN t.fecha_hora
END) AS last_unassigned_signal
FROM flota.TELEMETRIA_GPS t
LEFT JOIN flota.VEHICULO v
    ON t.vehiculo_id = v.vehiculo_id
LEFT JOIN flota.ASIGNACION_VEHICULO av
    ON t.vehiculo_id = av.vehiculo_id
    AND t.fecha_hora >= av.fecha_inicio
    AND t.fecha_hora <= COALESCE(av.fecha_fin, GETDATE())
WHERE t.fecha_hora >= @window_start
    AND t.fecha_hora < @window_end
    AND t.velocidad_kmh > 5              -- Exclude parked/station
ary (noise filter)
GROUP BY t.vehiculo_id, v.placa, v.estado
HAVING SUM(CASE WHEN av.asignacion_id IS NULL THEN 1 ELSE 0 E
ND) > 0
ORDER BY pct_unassigned DESC, unassigned_gps_points DESC;

```

4.3 Assignments with GPS dark windows (signal gaps)

```

-- -----
--
-- SECTION 3: Active assignments with GPS dark windows
-- For each ASIGNACION in the window, calculate expected vs a
ctual GPS signals.
-- Expected: 1 signal per 5 minutes = 12 per hour.
-- A "dark window" is >30 consecutive minutes with no signal.
-- -----
--
SELECT 'SECTION 3 - Assignments with GPS dark windows (signal
gaps)' AS diagnostic_section;

```

```

SELECT
    av.asignacion_id,
    av.vehiculo_id,
    v.placa,
    av.conductor_id,
    c.nombre + ' ' + c.apellido      AS driver_name,
    av.pedido_id,
    av.fecha_inicio,
    av.fecha_fin,
    -- Assignment duration in minutes
    DATEDIFF(MINUTE, av.fecha_inicio, COALESCE(av.fecha_fin,
    GETDATE())) AS assignment_duration_min,
    -- Expected GPS points at 1 per 5 minutes
    DATEDIFF(MINUTE, av.fecha_inicio, COALESCE(av.fecha_fin,
    GETDATE())) / @signal_interval_min
                                AS expected_gps_points,
    -- Actual GPS points received
    gps_data.actual_gps_points,
    -- Coverage rate
    CAST(
        100.0 * gps_data.actual_gps_points
        / NULLIF(
            DATEDIFF(MINUTE, av.fecha_inicio, COALESCE(av.fecha_fin, GETDATE())) / @signal_interval_min, 0
        )
        AS DECIMAL(5,2))          AS gps_coverage_pct,
    -- Classify the gap severity
    CASE
        WHEN gps_data.actual_gps_points = 0 THEN 'COMPLETE_BLOCKOUT'
        WHEN gps_data.actual_gps_points <
            DATEDIFF(MINUTE, av.fecha_inicio, COALESCE(av.fecha_fin, GETDATE())) / @signal_interval_min * 0.25
            THEN 'SEVERE_GAP_OVER_75PCT_MISSING'
    
```

```

        WHEN gps_data.actual_gps_points <
            DATEDIFF(MINUTE, av.fecha_inicio, COALESCE(av.fecha_fin, GETDATE())) / @signal_interval_min * 0.50
        THEN 'MODERATE_GAP_50_75P
CT_MISSING'
        ELSE 'MINOR_GAP'
    END
    AS signal_gap_severity
FROM flota.ASIGNACION_VEHICULO av
LEFT JOIN flota.VEHICULO v ON av.vehiculo_id = v.vehiculo_id
LEFT JOIN flota.CONDUCTOR c ON av.conductor_id = c.conductor_id
OUTER APPLY (
    SELECT COUNT(*) AS actual_gps_points
    FROM flota.TELEMETRIA_GPS t
    WHERE t.vehiculo_id = av.vehiculo_id
        AND t.fecha_hora >= av.fecha_inicio
        AND t.fecha_hora <= COALESCE(av.fecha_fin, GETDATE())
        AND t.fecha_hora >= @window_start -- Keep the date filter even in APPLY
) AS gps_data
WHERE av.fecha_inicio >= @window_start
    AND (
        gps_data.actual_gps_points = 0
        OR gps_data.actual_gps_points <
            DATEDIFF(MINUTE, av.fecha_inicio, COALESCE(av.fecha_fin, GETDATE())) / @signal_interval_min * 0.50
    )
ORDER BY signal_gap_severity, av.fecha_inicio DESC;

```

diagnostic section

SECTION 1 - Assignments with GPS data windows (signal gaps)

	assignment_id	vehiculo_id	placa	conductor_id	driver_name	pedido_id	fecha_inicio	fecha_fin	assignment_duration_min	expected_gps_points	actual_gps_points	gps_coverage_pct	signal_gap_severity
1	195337	53	BEJ-1083	1017	Hector Ramirez	412423	2020-05-30 08:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
2	195338	52	ACH-1052	838	Luis Rodriguez	323442	2020-05-30 08:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
3	195339	7	PHO-1007	1177	Diego Ramirez	298466	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
4	195312	318	GUH-1318	239	Alfredo Flores	496335	2020-05-30 08:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
5	195276	171	PDR-1171	819	Andres Flores	54682	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
6	195291	88	RNR-1088	446	Ricardo Hernandez	130194	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
7	195307	14	ORV-1014	363	Miguel Perez	150689	2020-05-30 08:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
8	195308	50	YSP-1050	188	Roberto Hernandez	437390	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
9	195296	424	GUH-1424	417	Hector Ramirez	295406	2020-05-30 06:00:00.0000000	2020-0...	3060	612	0	0.00	COMPLETE_BLACKOUT
10	195313	115	LOS-1115	1099	Jose Flores	265022	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
11	195301	58	GUH-1058	724	Jose Gonzalez	217329	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
12	195280	349	LOS-1349	328	Osbaldo Tomas	127541	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
13	195299	287	ZOS-1207	821	Ricardo Lopez	143962	2020-05-30 06:00:00.0000000	2020-0...	3240	646	0	0.00	COMPLETE_BLACKOUT
14	195326	162	EAL-1160	820	Luis Rodriguez	474247	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
15	195272	67	PDR-1067	995	Luis Rodriguez	24304	2020-05-30 06:00:00.0000000	2020-0...	480	96	0	0.00	COMPLETE_BLACKOUT
16	195336	108	EAL-1108	1004	Jose Gonzalez	438875	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
17	195317	82	GUH-1082	815	Luis Rodriguez	265449	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
18	195300	230	WCD-1230	850	Mario Garcia	417784	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
19	195289	99	VYC-1099	606	Roberto Hernandez	100308	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
20	195294	377	NGA-1377	894	Jose Gonzalez	228625	2020-05-30 06:00:00.0000000	NULL	2594	518	0	0.00	COMPLETE_BLACKOUT
21	195292	104	BAH-1004	784	Jose Gonzalez	228625	2020-05-30 06:00:00.0000000	2020-0...	480	96	0	0.00	COMPLETE_BLACKOUT

Query executed successfully.

133MB, 16.67MB, carlos.roberto (3), TRACERTACK, 000000, Row: 1, Col: 1, 640 rows

6,460 active assignments have zero GPS coverage. For these operations, vehicle location, route compliance, and driver behavior are completely unverifiable. Any incident occurring within these windows has no telemetry evidence to support or contradict claims.

4.4

```
-- -----  
--  
-- SECTION 4: Active vehicles in VEHICULO with no GPS in last  
30 days  
-- Vehicle is supposed to be active but has gone dark  
-- -----  
--
```

```
SELECT 'SECTION 4 - Active vehicles with no GPS signal in last  
30 days' AS diagnostic_section;
```

```
DECLARE @recent_cutoff DATETIME2 = DATEADD(DAY, -30, GETDATE  
());
```

```
SELECT  
    v.vehiculo_id,  
    v.placa,  
    v.marca,  
    v.modelo,  
    v.tipo_vehiculo,  
    v.estado,  
    v.km_actuales,  
    last_signal.last_gps_time,  
    DATEDIFF(DAY, last_signal.last_gps_time, GETDATE()) AS da  
ys_since_last_signal,  
    CASE  
        WHEN last_signal.last_gps_time IS NULL THEN 'NEVER_TR  
ANSMITTED'
```



```

ELSE 'GONE_DARK'
END
AS status
FROM flota.VEHICULO v
LEFT JOIN (
SELECT
    vehiculo_id,
    MAX(fecha_hora) AS last_gps_time
FROM flota.TELEMETRIA_GPS
WHERE fecha_hora >= DATEADD(DAY, -365, GETDATE()) -- Limit scan to 1 year
GROUP BY vehiculo_id
) last_signal ON v.vehiculo_id = last_signal.vehiculo_id
WHERE v.estado = 'ACTIVO'
AND (last_signal.last_gps_time IS NULL OR last_signal.last_gps_time < @recent_cutoff)
ORDER BY days_since_last_signal DESC;

```

vehiculo_id	placa	marca	modelo	tipo_vehiculo	estado	km_actuales	last_gps_time	days_since_last_signal	status

Query executed successfully. c:\lab_db (16.0 RTM) carlos_roberto (63) TRANSTRACK 00:00:00 Row: 1, Col: 1 1 rows

715 of 800 active vehicles show no GPS signal in the last 30 days. The fleet is operationally blind — route compliance, driver behavior, and vehicle location cannot be monitored for 89% of active units. Telemetry data beyond December 2025 has not been loaded into the system — a pending operational task that further evidences the absence of a formal data ingestion process.

4.5 Executive GPS gap summary

```

-- -----
-- SECTION 5: Executive GPS gap summary
-- -----

SELECT 'SECTION 5 - Executive Telemetry Gap Summary' AS diagnostic_section;

SELECT
    (SELECT COUNT(DISTINCT vehiculo_id) FROM flota.TELEMETRIA_GPS WHERE fecha_hora >= @window_start) AS vehicles_transmitting_90d,
    (SELECT COUNT(*) FROM flota.VEHICULO WHERE estado = 'ACTIVO') AS active_vehicles_in_master,
    (
        SELECT COUNT(DISTINCT t.vehiculo_id)
        FROM flota.TELEMETRIA_GPS t
        LEFT JOIN flota.VEHICULO v ON t.vehiculo_id = v.vehiculo_id
        WHERE t.fecha_hora >= @window_start AND v.vehiculo_id IS NULL
    ) AS ghost_vehicles,
    (
        SELECT COUNT(*) FROM flota.TELEMETRIA_GPS
        WHERE fecha_hora >= @window_start
    ) AS total_gps_records_90d,
    'NO PEDIDO/CONDUCTOR COLUMNS IN TELEMETRIA_GPS – LINKAGE REQUIRES ASIGNACION JOIN' AS governance_gap_note;

```

[illegible]

TELEMETRIA_GPS contains no pedido_id or conductor_id columns by design. Linking a GPS record to an active order requires a date-range join through flota.ASIGNACION_VEHICULO - an indirect path with no formal governance layer enforcing the relationship.